Martingale Pricing, Hedging, and No-Arbitrage

Lecture 7 Notes — From Martingale Prices to Replication and Market Completeness

Goals and logical structure

Lecture 6 constructed the pricing measure. Lecture 7 follows one chain:

risk-neutral stock dynamics $\longrightarrow$ discounted self-financing gains and $\mathbb{Q}$-prices
$\longrightarrow$ dividend-paying call replication under $\mathbb{P}$
$\longrightarrow$ dividend-adjusted pricing and a new numeraire
$\longrightarrow$ martingale hedges and two-risk spanning
$\longrightarrow$ completeness and path dependence
$\longrightarrow$ Itô, Girsanov, reflection, and stopping-time revision.

The recurring discipline is to separate four questions: which numeraire is used, under which measure the discounted gains are martingales, whether the claim is replicable, and whether the relevant local martingales are true martingales.

1 The Risk-Neutral Starting Point

1.1 The five-line bridge from Lecture 6

For the constant-coefficient stock model under $\mathbb{P}$, let $\lambda = (\mu - r)/\sigma$. The preceding lecture ended with the chain

$$Z_t = e^{-\lambda W_t^{\mathbb{P}} - \lambda^2 t/2}, \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T, \qquad W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \lambda t,$$

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \mathrm{d}(e^{-rt}S_t) = \sigma e^{-rt}S_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

The density changes probabilities; the last expression identifies a stochastic integral. Its true-martingale status needs a separate integrability argument.

For now, use the money-market account $B^{(0)}$ as the numeraire. Under its equivalent martingale measure, discounted traded prices or total gains are $\mathbb{Q}$-local martingales; under the conditions used for valuation, they are true martingales. We change numeraire after treating dividends.

1.2 Why the discounted stock is a local martingale

For a non-dividend-paying stock under the risk-neutral measure, suppose

$$\mathrm{d}S_t = r_t S_t\,\mathrm{d}t + \sigma_t S_t\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad D_t = \exp\left(-\int_0^t r_u\,\mathrm{d}u\right).$$

The product rule removes the drift:

$$\boxed{D_t S_t = S_0 + \int_0^t D_u \sigma_u S_u\,\mathrm{d}W_u^{\mathbb{Q}}.}$$

A well-defined stochastic integral with respect to Brownian motion is a local martingale. This statement by itself does not assert that the discounted price is a true martingale; that requires an additional integrability argument.

The stock is now a local martingale after discounting. We next show how self-financing gains lead to a conditional-expectation price whenever a portfolio replicates a payoff.

2 Self-Financing Leads to Martingale Prices

We first derive the martingale valuation identity for any replicating self-financing strategy. We will then construct such a strategy explicitly for a call on a dividend-paying stock.

2.1 Discounted self-financing gains

Let the money-market account be

$$B_t^{(0)} = \exp\left(\int_0^t r_u \, \mathrm{d}u\right), \qquad D_t = \frac{1}{B_t^{(0)}}.$$

For this derivation, suppose the risky assets pay no dividends. Consider $S^1, \dots, S^m$ and a predictable strategy $\phi = (\phi^0, \phi^1, \dots, \phi^m)$ with wealth

$$X_t = \phi_t^0 B_t^{(0)} + \sum_{i=1}^m \phi_t^i S_t^i.$$

The strategy is self-financing when

$$\mathrm{d}X_t = \phi_t^0 \, \mathrm{d}B_t^{(0)} + \sum_{i=1}^m \phi_t^i \, \mathrm{d}S_t^i.$$

Define discounted prices

$$\bar{X}_t = D_t X_t, \qquad \bar{S}_t^i = D_t S_t^i.$$

Proposition 2.1 (Discounted gain identity)**.** For a self-financing strategy,

$$\boxed{\mathrm{d}\bar{X}_t = \sum_{i=1}^m \phi_t^i \, \mathrm{d}\bar{S}_t^i.}$$

Equivalently,

$$\bar{X}_t = \bar{X}_0 + \sum_{i=1}^m \int_0^t \phi_u^i \, \mathrm{d}\bar{S}_u^i.$$

Proof. Since D has finite variation,

$$\mathrm{d}(D_t X_t) = D_t \, \mathrm{d}X_t + X_t \, \mathrm{d}D_t, \qquad \mathrm{d}D_t = -r_t D_t \, \mathrm{d}t.$$

Use $\mathrm{d}B_t^{(0)} = r_t B_t^{(0)} \, \mathrm{d}t$ and $\phi_t^0 B_t^{(0)} = X_t - \sum_i \phi_t^i S_t^i$. The financing terms cancel, leaving precisely the discounted risky-asset gains. □

Exercise — recover the cancellation in one dimension

For $X_t = \Delta_t S_t + \beta_t B_t^{(0)}$, start from $\mathrm{d}X_t = \Delta_t \, \mathrm{d}S_t + \beta_t \, \mathrm{d}B_t^{(0)}$ and derive $\mathrm{d}(D_t X_t) = \Delta_t \, \mathrm{d}(D_t S_t)$.

Short solution

$$\begin{aligned} \mathrm{d}(DX) &= D\Delta \, \mathrm{d}S + D\beta r B^{(0)} \, \mathrm{d}t - rDX \, \mathrm{d}t \\ &= \Delta(D \, \mathrm{d}S - rDS \, \mathrm{d}t) = \Delta \, \mathrm{d}(DS). \end{aligned}$$

The equality uses $X = \Delta S + \beta B^{(0)}$. Discounting removes the cash account from the gain equation, not from the portfolio value.

2.2 The martingale valuation identity

Fix an equivalent martingale measure $\mathbb{Q}$ associated with the money-market numeraire. By definition, discounted traded gains are $\mathbb{Q}$-local martingales. Under the required admissibility and integrability assumptions, the discounted wealth of a self-financing strategy is a true martingale.

Theorem 2.2 (Pricing a replicable claim)**.** Let H be $\mathcal{F}_T$-measurable and suppose a self-financing strategy replicates it:

$$X_T = H.$$

If $\bar{X} = DX$ is a true $\mathbb{Q}$-martingale, then its value is

$$\boxed{V_t = X_t = B_t^{(0)}\mathbb{E}^{\mathbb{Q}}\left[\frac{H}{B_T^{(0)}}\middle|\mathcal{F}_t\right] = \frac{1}{D_t}\mathbb{E}^{\mathbb{Q}}[D_T H \mid \mathcal{F}_t].}$$

When rates are deterministic,

$$\boxed{V_t = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u\,\mathrm{d}u\right)H\middle|\mathcal{F}_t\right].}$$

Proof. The discounted gain identity makes $D_t X_t$ a stochastic integral with respect to discounted traded martingales. Once it is a true martingale,

$$D_t X_t = \mathbb{E}^{\mathbb{Q}}[D_T X_T \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[D_T H \mid \mathcal{F}_t].$$

Multiply by $D_t^{-1} = B_t^{(0)}$. □

Local martingale is not enough

No drift is an algebraic statement; pricing by conditional expectation is an integrability statement. A discounted self-financing wealth process is first a local martingale. Admissibility, square integrability, nonnegativity, or a problem-specific argument is needed before asserting $\bar{X}_t = \mathbb{E}^{\mathbb{Q}}[\bar{X}_T \mid \mathcal{F}_t]$.

Example — a unit payoff and a terminal share

Assume constant r and no dividends. For $H = 1$,

$$V_t = e^{-r(T-t)}.$$

For $H = S_T$,

$$V_t = B_t^{(0)}\mathbb{E}^{\mathbb{Q}}\left[\frac{S_T}{B_T^{(0)}}\middle|\mathcal{F}_t\right] = S_t,$$

because $S/B^{(0)}$ is a $\mathbb{Q}$-martingale. A payoff of one bank-account unit at T would instead cost $B_t^{(0)}$. These values can also be read directly from their replicating portfolios.

2.3 The lognormal law, digitals, and stock-weighted tails

For a replicable integrable payoff H, when discounted wealth is a true martingale, constant rates give

$$\boxed{V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t].}$$

Under $\mathbb{Q}$, conditional on $\mathcal{F}_t$, the Brownian increment $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}}$ is normal with mean zero and variance $\tau = T - t$. Hence

$$S_T = S_t \exp\Big((r - \tfrac{1}{2}\sigma^2)\tau + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})\Big).$$

Define

$$d_2 = \frac{\log(S_t/K) + (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_1 = d_2 + \sigma\sqrt{\tau}.$$

A cash-or-nothing digital paying $\mathbf{1}_{\{S_T > K\}}$ has value

$$\boxed{D_t = e^{-r\tau}\mathbb{Q}(S_T > K \mid \mathcal{F}_t) = e^{-r\tau}\Phi(d_2).}$$

A stock-or-nothing claim paying $S_T\mathbf{1}_{\{S_T > K\}}$ has value

$$\boxed{A_t = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T > K\}} \mid \mathcal{F}_t] = S_t\Phi(d_1).}$$

The second tail weights each outcome by its stock price. Completing the square in the normal density shifts the standardized cutoff by $\sigma\sqrt{\tau}$, which is why it uses d_1 instead of d_2. Combining the two claims gives $C_t = A_t - KD_t$. The dividend-paying replication argument below will give the corresponding result for $q > 0$.

2.4 Forwards and call decomposition

Example — a forward payoff

For $H = S_T - K$ and a non-dividend-paying stock,

$$\begin{aligned} V_t &= e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[S_T - K \mid \mathcal{F}_t] \\ &= S_t - Ke^{-r(T-t)}. \end{aligned}$$

The delivery price that makes the contract worth zero is therefore

$$\boxed{K_t = S_t e^{r(T-t)}.}$$

This is both a martingale calculation and the law-of-one-price value of *one share minus a zero-coupon payment.*

For a call,

$$(S_T - K)^+ = S_T \mathbf{1}_{\{S_T > K\}} - K\mathbf{1}_{\{S_T > K\}},$$

so

$$C_t = e^{-r(T-t)}\left(\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T > K\}} \mid \mathcal{F}_t] - K\mathbb{Q}(S_T > K \mid \mathcal{F}_t)\right).$$

The first term is stock weighted; the second is a discounted cash tail. This explains structurally why the Black–Scholes formula contains two different normal terms.

Exercise — value before formula

Let $H = aS_T + bB_T^{(0)}$, where a, b are constants. Price the claim without computing any distribution.

Short solution

The claim is already replicated by a shares and b bank-account units:

$$\boxed{V_t = aS_t + bB_t^{(0)}.}$$

The martingale identity gives the same result by linearity.

3 Dividend Yield and Risk-Neutral Pricing

3.1 Continuous dividends and total gains

If the stock pays a continuous dividend yield q_t, one share earns the total gain

$$\boxed{\mathrm{d}G_t = \mathrm{d}S_t + q_t S_t \,\mathrm{d}t.}$$

The ex-dividend price S_t excludes cash distributions. A holder of Δ_t shares therefore earns $\Delta_t \,\mathrm{d}G_t$, not merely $\Delta_t \,\mathrm{d}S_t$. This distinction matters both for replication and for the martingale condition.

3.2 Replicate a dividend-paying call under the historical measure

Assume constant r, q, σ, with $\sigma > 0$, and let μ be the historical *ex-dividend* price drift:

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{P}}, \qquad \mathrm{d}B_t = rB_t \,\mathrm{d}t.$$

Seek a smooth call value $V_t = v(t, S_t)$ with $v(T, s) = (s - K)^+$. Itô's formula yields

$$\mathrm{d}v = \left(v_t + \mu S_t v_s + \tfrac{1}{2}\sigma^2 S_t^2 v_{ss}\right)\,\mathrm{d}t + \sigma S_t v_s \,\mathrm{d}W_t^{\mathbb{P}}.$$

A self-financing portfolio holds Δ_t shares and invests $v(t,S_t)-\Delta_t S_t$ in the bank account. Its gain includes the stock's dividend cash flow:

$$\boxed{\mathrm{d}V_t=\Delta_t(\,\mathrm{d}S_t+qS_t\,\mathrm{d}t)+r(V_t-\Delta_tS_t)\,\mathrm{d}t.}$$

Rebalancing the holdings is financed within the portfolio; no external cash is injected. Matching Brownian coefficients gives $\Delta_t=v_s(t,S_t)$. Matching the drifts then gives

$$v_t+\mu sv_s+\tfrac{1}{2}\sigma^2s^2v_{ss}=\mu sv_s+qsv_s+r(v-sv_s).$$

The historical drift cancels. The dividend-adjusted Black–Scholes replication PDE is

$$\boxed{v_t+\tfrac{1}{2}\sigma^2s^2v_{ss}+(r-q)sv_s-rv=0,\qquad v(T,s)=(s-K)^+.}$$

A sufficiently regular solution with an admissible self-financing hedge replicates the call. No-arbitrage therefore identifies its price with $v(t,S_t)$.

3.3 Prepaid shares, the pricing measure, and the call formula

For constant q, a prepaid forward delivering one share at T without interim dividends is worth $S_te^{-q(T-t)}$. Buy $e^{-q(T-t)}$ shares now and reinvest the dividends: the holding reaches one share at T. Consequently,

$$\boxed{F_{t,T}=S_te^{(r-q)(T-t)},\qquad C_t-P_t=S_te^{-q(T-t)}-Ke^{-r(T-t)}.}$$

The measure associated with the money-market numeraire has Brownian shift

$$\lambda=\frac{\mu+q-r}{\sigma},\qquad Z_t=\exp(-\lambda W_t^{\mathbb{P}}-\tfrac{1}{2}\lambda^2t),\qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T}=Z_T.$$

Then $W_t^{\mathbb{Q}}=W_t^{\mathbb{P}}+\lambda t$ is Brownian under $\mathbb{Q}$, and

$$\boxed{\frac{\mathrm{d}S_t}{S_t}=(r-q)\,\mathrm{d}t+\sigma\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

In particular,

$$\mathrm{d}(D_tS_t)+D_tqS_t\,\mathrm{d}t=D_t\sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}},\qquad D_t=e^{-rt}.$$

Thus discounted *total gains* are driftless, while the discounted ex-dividend price alone need not be a martingale.

The Feynman–Kac representation of the replication PDE is

$$v(t,s)=e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(S_T-K)^+\mid S_t=s],\qquad \tau=T-t.$$

Conditional on $S_t=s$, write

$$S_T=s\exp\!\big((r-q-\tfrac{1}{2}\sigma^2)\tau+\sigma\sqrt{\tau}Z\big),\qquad Z\sim\mathcal{N}(0,1).$$

Define

$$d_2^q=\frac{\log(s/K)+(r-q-\sigma^2/2)\tau}{\sigma\sqrt{\tau}},\qquad d_1^q=d_2^q+\sigma\sqrt{\tau}.$$

The Gaussian cash and stock tails are

$$\mathbb{Q}(S_T>K\mid S_t=s)=\Phi(d_2^q),\qquad e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T>K\}}\mid S_t=s]=se^{-q\tau}\Phi(d_1^q).$$

Using $(S_T-K)^+=S_T\mathbf{1}_{\{S_T>K\}}-K\mathbf{1}_{\{S_T>K\}}$ yields the replicating value

$$\boxed{C_t=S_te^{-q\tau}\Phi(d_1^q)-Ke^{-r\tau}\Phi(d_2^q).}$$

The stock holding is the derivative of this solution:

$$\boxed{\Delta_t=v_s(t,S_t)=e^{-q\tau}\Phi(d_1^q),}\qquad C_t-\Delta_tS_t=-Ke^{-r\tau}\Phi(d_2^q).$$

When differentiating, the terms involving the normal density cancel because $se^{-q\tau}\varphi(d_1^q)=Ke^{-r\tau}\varphi(d_2^q)$. This price solves the PDE with the call terminal payoff, and the displayed holdings implement the replication.

Exercise — spot the wrong martingale

Under $\mathbb{Q}$, suppose $\mathrm{d}S_t/S_t = (r-q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}$ with $q > 0$. Is $e^{-rt}S_t$ a martingale? Identify the correct martingale object.

Short solution

$$\frac{\mathrm{d}(e^{-rt}S_t)}{e^{-rt}S_t} = -q\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}},$$

so the discounted ex-dividend price is not a martingale. The discounted cum-dividend gain is driftless. With constant q, the reinvested process $e^{qt}S_t$, after discounting, is a local martingale.

4 Changing the Numeraire

4.1 A general numeraire

Let N be a strictly positive traded numeraire. Under its associated measure $\mathbb{Q}^N$, prices expressed in units of N are martingales. A replicable payoff therefore satisfies

$$\boxed{V_t = N_t \mathbb{E}^{\mathbb{Q}^N}\left[\frac{H}{N_T}\middle|\mathcal{F}_t\right].}$$

Changing numeraire changes both units and measure. It does not change the economic value V_t.

Example — choose the numeraire that is already in the payoff

For $H = cN_T$,

$$V_t = N_t \mathbb{E}^{\mathbb{Q}^N}[c \mid \mathcal{F}_t] = cN_t.$$

The best numeraire can turn a random payoff into a constant ratio.

Three formulas that should not be mixed

money-market numeraire:	$V_t = B_t^{(0)}\mathbb{E}^{\mathbb{Q}}[H/B_T^{(0)} \mid \mathcal{F}_t],$
constant rate:	$V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t],$
general numeraire:	$V_t = N_t\mathbb{E}^{\mathbb{Q}^N}[H/N_T \mid \mathcal{F}_t].$

The numeraire, measure, and discount factor are one matched package.

5 From Prices to Hedges: Martingale Representation

5.1 The representation theorem

The conditional expectation gives a price process, but it does not yet state which assets to hold. Brownian martingale representation provides the missing link.

Theorem 5.1 (Martingale Representation Theorem)**.** Let $(\mathcal{F}_t)$ be the augmented filtration generated by a d-dimensional $\mathbb{Q}$-Brownian motion $W^{\mathbb{Q}}$. If M is a square-integrable $\mathbb{Q}$-martingale, then there is a predictable process $\psi_t \in \mathbb{R}^d$ such that

$$\boxed{M_t = M_0 + \int_0^t \psi_u^{\top}\,\mathrm{d}W_u^{\mathbb{Q}}, \qquad \mathbb{E}^{\mathbb{Q}}\int_0^T \|\psi_u\|^2\,\mathrm{d}u < \infty.}$$

The integrand is unique up to $\mathrm{d}t \otimes \mathrm{d}\mathbb{Q}$-null sets.

For a claim H, take

$$M_t = \mathbb{E}^{\mathbb{Q}}[D_T H \mid \mathcal{F}_t] = D_t V_t.$$

The coefficient ψ_t records the exposure of the discounted claim to each Brownian shock.

Example — linear and quadratic terminal shocks

For $H = W_T^{\mathbb{Q}}$,

$$\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t] = W_t^{\mathbb{Q}} = \int_0^t 1 \, \mathrm{d}W_u^{\mathbb{Q}}.$$

For $H = (W_T^{\mathbb{Q}})^2$,

$$\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t] = (W_t^{\mathbb{Q}})^2 + (T - t),$$

and Itô's formula gives

$$\mathrm{d}\big((W_t^{\mathbb{Q}})^2 + T - t\big) = 2W_t^{\mathbb{Q}} \, \mathrm{d}W_t^{\mathbb{Q}}.$$

Thus the representation coefficient is 1 in the first example and $2W_t^{\mathbb{Q}}$ in the second.

Exercise — represent a cubic terminal shock

Find $M_t = \mathbb{E}^{\mathbb{Q}}[(W_T^{\mathbb{Q}})^3 \mid \mathcal{F}_t]$ and its representation coefficient.

Short solution

Write $W_T^{\mathbb{Q}} = W_t^{\mathbb{Q}} + Z$, where $Z \sim \mathcal{N}(0, T - t)$ is independent of $\mathcal{F}_t$. Then

$$M_t = (W_t^{\mathbb{Q}})^3 + 3(T - t)W_t^{\mathbb{Q}}.$$

Applying Itô's formula,

$$\boxed{\mathrm{d}M_t = 3\big((W_t^{\mathbb{Q}})^2 + T - t\big) \, \mathrm{d}W_t^{\mathbb{Q}}.}$$

The coefficient contains both the risk already observed and the variance still to come.

5.2 One traded asset

Suppose the discounted stock satisfies

$$\mathrm{d}\bar{S}_t = \gamma_t \, \mathrm{d}W_t^{\mathbb{Q}}, \qquad \gamma_t \neq 0,$$

and the discounted claim martingale satisfies

$$\mathrm{d}\bar{V}_t = \psi_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

A self-financing hedge holding Δ_t shares obeys

$$\mathrm{d}\bar{V}_t = \Delta_t \, \mathrm{d}\bar{S}_t = \Delta_t \gamma_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

Matching Brownian coefficients gives

$$\boxed{\Delta_t = \frac{\psi_t}{\gamma_t}.}$$

In the Black–Scholes model, $\gamma_t = \sigma_t \bar{S}_t$, so

$$\boxed{\Delta_t = \frac{\psi_t}{\sigma_t \bar{S}_t}.}$$

If $V_t = v(t, S_t)$ is smooth and Markov, Itô's formula yields

$$\psi_t = D_t \sigma_t S_t v_s(t, S_t),$$

and therefore

$$\boxed{\Delta_t = v_s(t, S_t).}$$

The PDE delta and the martingale-representation hedge are the same object written in state coordinates and Brownian coordinates, respectively.

6 Two Brownian Risks and Market Completeness

6.1 A call with an untraded sector state

Under the historical measure, interpret W^1 as an AI-sector shock and W^2 as a broad market shock. For classroom calculations suppose they are independent. This is a simplifying modeling assumption: empirical sector and market shocks may be correlated. Ask whether independence is plausible and how correlation would affect the two risk directions.

Let the traded AI stock S and an untraded sector state Y satisfy

$$\frac{\mathrm{d}S_t}{S_t} = \mu(Y_t)\,\mathrm{d}t + \sigma_1(Y_t)\,\mathrm{d}W_t^1 + \sigma_2\,\mathrm{d}W_t^2, \qquad \mathrm{d}Y_t = a(Y_t)\,\mathrm{d}t + b(Y_t)\,\mathrm{d}W_t^1,$$

with $\sigma_2 \neq 0$. The sector state changes the stock's volatility. For a vanilla call $H = (S_T - K)^+$, its value is generally $v(t, S_t, Y_t)$. Itô's formula gives Brownian exposure

$$\big(S\sigma_1 v_s + b v_y,\ S\sigma_2 v_s\big).$$

Holding Δ shares of S produces only $\big(\Delta S\sigma_1, \Delta S\sigma_2\big)$. The common-market component forces $\Delta = v_s$, leaving $b v_y$ of sector exposure uncovered whenever it is nonzero.

Why a second state is needed

If the stock coefficients were constant and the call value depended only on S_t, then $v_y = 0$. Delta hedging that call would still work even though the stock had two Brownian terms. The example exposes incompleteness through the untraded sector state, not simply by counting two noises in the stock equation.

6.2 A second traded product and the rank condition

Suppose a second traded asset R has dynamics under $\mathbb{P}$

$$\frac{\mathrm{d}R_t}{R_t} = \nu(Y_t)\,\mathrm{d}t + \eta_1(Y_t)\,\mathrm{d}W_t^1 + \eta_2(Y_t)\,\mathrm{d}W_t^2.$$

The holdings (Δ^S, Δ^R) match the call's two shocks when

$$\begin{pmatrix} S\sigma_1 & R\eta_1 \\ S\sigma_2 & R\eta_2 \end{pmatrix} \begin{pmatrix} \Delta^S \\ \Delta^R \end{pmatrix} = \begin{pmatrix} S\sigma_1 v_s + b v_y \\ S\sigma_2 v_s \end{pmatrix}.$$

If $\sigma_1\eta_2 - \sigma_2\eta_1 \neq 0$, the exposure matrix has full rank and this system has a unique solution. Under the usual integrability and admissibility conditions, full rank throughout the modeled Brownian filtration is the spanning mechanism behind market completeness: every suitable terminal claim can be replicated.

6.3 The matching martingale measure

The market-price-of-risk vector is determined by both traded assets:

$$\begin{pmatrix} \sigma_1 & \sigma_2 \\ \eta_1 & \eta_2 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} \mu - r \\ \nu - r \end{pmatrix}.$$

Write $W^{\mathbb{P}} = (W^{1,\mathbb{P}}, W^{2,\mathbb{P}})^\top$ and $\lambda = (\lambda_1, \lambda_2)^\top$. The density process and the terminal change of measure are

$$\boxed{Z_t = \exp\!\left(-\int_0^t \lambda_u^\top\,\mathrm{d}W_u^{\mathbb{P}} - \frac{1}{2}\int_0^t \|\lambda_u\|^2\,\mathrm{d}u\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.}$$

When Z is a true martingale, Girsanov gives $\mathrm{d}W_t^{i,\mathbb{Q}} = \mathrm{d}W_t^{i,\mathbb{P}} + \lambda_i(t)\,\mathrm{d}t$. Both discounted traded asset prices are then $\mathbb{Q}$-local martingales. For the call, subject to the valuation martingale conditions,

$$V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ \mid \mathcal{F}_t].$$

The martingale representation of the discounted claim gives two Brownian coefficients. Matching them with the two traded exposures recovers the same hedge as replication under $\mathbb{P}$.

6.4 Several assets: hedging is a linear system

Suppose m discounted risky assets are driven by d Brownian factors:

$$\mathrm{d}\bar{S}_t = \Gamma_t \,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \Gamma_t \in \mathbb{R}^{m\times d}.$$

If the claim has representation

$$\mathrm{d}\bar{V}_t = \psi_t^{\top} \,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \psi_t \in \mathbb{R}^{d},$$

then holdings $\phi_t \in \mathbb{R}^m$ replicate its Brownian exposure precisely when

$$\boxed{\Gamma_t^{\top}\phi_t = \psi_t.}$$

Exercise — solve a two-factor hedge

Let

$$\Gamma = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}, \qquad \psi = \begin{pmatrix} 4 \\ 6 \end{pmatrix}.$$

Find the asset holdings $\phi = (\phi_1, \phi_2)^{\top}$.

Short solution

The hedge equation is

$$\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}.$$

Hence $\phi_2 = 2$ and $\phi_1 = 1$. Both Brownian shocks are spanned.

Representation versus implementation

The MRT says every square-integrable Brownian martingale has a Brownian exposure ψ. Market completeness asks whether traded assets can implement that exposure by solving $\Gamma^{\top}\phi = \psi$. These are different statements.

6.5 Completeness is a spanning condition

Definition 6.1 (Complete market)**.** A market is complete on $[0, T]$ if every claim in the chosen admissible class can be replicated by a self-financing trading strategy.

In a Brownian diffusion market, completeness is essentially a spanning condition. If the filtration has d independent Brownian risks and

$$\operatorname{rank}(\Gamma_t) = d \quad \mathrm{d}t \otimes \mathrm{d}\mathbb{Q}\text{-almost everywhere},$$

then every Brownian exposure $\psi_t \in \mathbb{R}^d$ can be matched. Full column rank requires at least d sufficiently independent risky assets.

The two-asset example above has full rank. Removing a spanning asset leaves a Brownian risk that traded prices do not constrain.

6.6 Incomplete markets and several martingale measures

Consider under $\mathbb{P}$

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^1,$$

while the filtration is generated by independent Brownian motions (W^1, W^2). To make the discounted stock driftless, the first Brownian shift is fixed:

$$\lambda_1 = \frac{\mu - r}{\sigma}.$$

The shift of W^2 is not constrained by any traded asset. Different choices therefore produce different equivalent martingale measures. For a constant free shift η, one such density is

$$Z_T^{\eta} = \exp\!\left(-\lambda_1 W_T^1 - \eta W_T^2 - \tfrac{1}{2}(\lambda_1^2 + \eta^2)T\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}^{\eta}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T^{\eta}.$$

Example — a nontraded claim reveals price ambiguity

Let $H = (W_T^2)^2$. Under a pricing measure indexed by a constant η, write

$$W_t^{2,\mathbb{Q}^\eta} = W_t^2 + \eta t.$$

Then $W_T^2 = W_T^{2,\mathbb{Q}^\eta} - \eta T$, and

$$\mathbb{E}^{\mathbb{Q}^\eta}[H] = T + \eta^2 T^2.$$

The stock does not constrain η, so martingale measures disagree on the expectation of this nonreplicable claim. No-arbitrage alone does not select one price.

Several distinctions matter

- Multiple EMMs signal incompleteness; they are not themselves an arbitrage.
- A particular claim can have the same expectation under every EMM even when the market is incomplete.
- Redundant assets can make portfolio holdings nonunique even when attainable payoffs have unique prices.
- Completeness is about spanning payoffs, not about counting portfolio labels.

6.7 Test the rank criterion

Exercise — rank, hedge, and ambiguity

Classify each case for two Brownian factors:

$$\Gamma_1 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \Gamma_2 = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}.$$

Which matrix spans every exposure? For $\psi = (1, -1)^\top$, does a hedge exist in each case?

Short solution

Γ_1 has rank two, so every exposure is spanned; the hedge solves $\Gamma_1^\top \phi = \psi$, giving $\phi = (1, -1/2)^\top$. The matrix Γ_2 has rank one. The range of $\Gamma_2^\top$ is generated by $(1, 1)^\top$, while $(1, -1)^\top$ is not in that space. Equivalently, $\Gamma_2^\top \phi = (\phi_1 + 2\phi_2, \phi_1 + 2\phi_2)^\top$, so the second market cannot hedge this exposure.

7 Asian Options and State Augmentation

We now return to the complete one-factor Black–Scholes model. Completeness settles price uniqueness; path dependence raises a different question about which state variables are needed to evaluate that price.

7.1 Markov structure is optional

If $H = g(S_T)$ and S is Markov under $\mathbb{Q}$, the conditional expectation can often be written as $V_t = v(t, S_t)$. This makes PDE methods available. The pricing identity itself is more general:

$$V_t = B_t^{(0)} \mathbb{E}^{\mathbb{Q}} \left[\frac{H}{B_T^{(0)}} \middle| \mathcal{F}_t \right]$$

still makes sense when H depends on the entire path or when the state is high-dimensional. Markov structure simplifies the representation of the conditional expectation; it is not the source of the pricing principle.

7.2 Path dependence changes the state, not the pricing principle

A fixed-strike arithmetic-average call has payoff

$$H = \left(\frac{1}{T} \int_0^T S_u \, \mathrm{d}u - K \right)^+ .$$

Two stock paths can have the same endpoint and different averages, so S_t alone does not contain enough information to value the claim. Nevertheless,

$$\boxed{V_t = B_t^{(0)} \mathbb{E}^{\mathbb{Q}} \left[\frac{H}{B_T^{(0)}} \middle| \mathcal{F}_t \right]}$$

remains valid. Path dependence complicates the state and computation, not the martingale valuation identity.

Exercise — same endpoint, different claim value

On $[0, 1]$, compare the deterministic paths

$$S_u^{(1)} = 100, \qquad S_u^{(2)} = \begin{cases} 140, & u < 3/4, \\ 60, & 3/4 \le u < 1, \\ 100, & u = 1. \end{cases}$$

They have the same endpoint. Compare their arithmetic averages and explain why a terminal-state function $v(t, S_t)$ cannot distinguish them.

Short solution

The first average is 100. Ignoring the single endpoint, which has zero Lebesgue measure, the second average is

$$\frac{3}{4}(140) + \frac{1}{4}(60) = 120.$$

For a strike between 100 and 120, the two paths generate different Asian payoffs despite having the same terminal stock value. Endpoint information alone cannot determine the running integral.

7.3 The running integral restores Markov structure

Introduce

$$\boxed{A_t = \int_0^t S_u \, \mathrm{d}u, \qquad \mathrm{d}A_t = S_t \, \mathrm{d}t.}$$

Then

$$H = \left(\frac{A_T}{T} - K \right)^+ ,$$

and the enlarged state is (S_t, A_t). Under constant r, q, σ,

$$\frac{\mathrm{d}S_t}{S_t} = (r - q) \, \mathrm{d}t + \sigma \, \mathrm{d}W_t^{\mathbb{Q}}, \qquad \mathrm{d}A_t = S_t \, \mathrm{d}t.$$

The price may therefore be written

$$v(t, s, a) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} \left[\left(\frac{A_T}{T} - K \right)^+ \middle| S_t = s, \ A_t = a \right].$$

Proposition 7.1 (Two-state Asian PDE)**.** Under the preceding assumptions, a sufficiently smooth value function solves

$$\boxed{v_t + (r - q) s v_s + s v_a + \frac{1}{2} \sigma^2 s^2 v_{ss} - r v = 0,}$$

with terminal condition

$$\boxed{v(T, s, a) = \left(\frac{a}{T} - K \right)^+ .}$$

Proof. Apply the two-dimensional Itô formula:

$$\mathrm{d}v = \left(v_t + (r-q)sv_s + sv_a + \frac{1}{2}\sigma^2 s^2 v_{ss}\right)\mathrm{d}t + \sigma s v_s \,\mathrm{d}W_t^{\mathbb{Q}}.$$

Discounting and setting the drift to zero gives the PDE. There is no v_{aa} term because A has finite variation and no direct Brownian shock. □

Do not invent diffusion terms

The state has two coordinates but only one Brownian driver. Since $\mathrm{d}A_t = S_t\,\mathrm{d}t$,

$$\mathrm{d}[A]_t = 0, \qquad \mathrm{d}[S,A]_t = 0.$$

State dimension and noise dimension are not the same thing.

7.4 Arithmetic, geometric, and discrete monitoring

For a geometric-average option,

$$G_T = \exp\left(\frac{1}{T}\int_0^T \log S_u\,\mathrm{d}u\right).$$

In Black–Scholes, $\log G_T$ is Gaussian because it is a linear functional of Brownian motion. This often yields a closed form. The arithmetic average is an integral of a lognormal process and generally has no elementary distribution.

For monitoring dates $t_1 < \cdots < t_n$, define

$$A_k = \sum_{j=1}^{k} S_{t_j}.$$

Then

$$A_{k+1} = A_k + S_{t_{k+1}}, \qquad H = \left(\frac{A_n}{n} - K\right)^+.$$

Between fixing dates, A_k is constant; at a fixing date, it jumps by the new observation. Dynamic programming therefore uses the state (S_{t_k}, A_k, k).

Example — Monte Carlo workflow for an arithmetic Asian call

For M simulated paths:

1. simulate the stock under $\mathbb{Q}$;
2. update the running sum or time integral along each path;
3. compute $H^{(m)}$ at maturity;
4. estimate
$$\widehat{V}_0 = e^{-rT}\frac{1}{M}\sum_{m=1}^{M} H^{(m)};$$
5. report a standard error
$$e^{-rT}\frac{\widehat{\mathrm{sd}}(H)}{\sqrt{M}}.$$

Path dependence is computationally natural for simulation because the full path is generated anyway.

Exercise — select the minimal state

For each payoff, propose a Markov state augmentation:

$$\max_{0\le u\le T} S_u - K, \qquad \mathbf{1}_{\{\min_{0\le u\le T} S_u > L\}}(S_T - K)^+, \qquad \left(\frac{1}{T}\int_0^T S_u\,\mathrm{d}u - K\right)^+.$$

Short solution

Track respectively the running maximum $M_t = \max_{u\le t} S_u$, the running minimum $m_t = \min_{u\le t} S_u$ or a barrier-survival indicator, and the running integral $A_t = \int_0^t S_u\,\mathrm{d}u$. The state should summarize precisely the past information that the remaining payoff needs.

8 Synthesis: Choosing the Right Route

8.1 Choose the right concept before calculating

Exercise — diagnose six situations

Match each situation with the main concept it tests:

1. discounted traded prices retain a nonzero drift under a candidate measure;
2. two portfolios have identical terminal cashflows but different costs;
3. a claim's Brownian exposure is outside the span of $\Gamma_t^\top$;
4. exercise is allowed before maturity;
5. a payoff depends on $\int_0^T S_u\,\mathrm{d}u$;
6. several equivalent martingale measures price a nontraded risk differently.

Short solution

1. the measure is not an EMM for the chosen numeraire;
2. law of one price and an arbitrage opportunity;
3. nonreplicability and market incompleteness;
4. optimal stopping and the Snell envelope;
5. path dependence and state augmentation;
6. incomplete-market price ambiguity.

8.2 A complete martingale-pricing drill

Exercise — from payoff to hedge

Suppose r is constant and

$$\mathrm{d}\bar{S}_t = \sigma \bar{S}_t\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \sigma > 0.$$

Let $H \in L^2(\mathbb{Q})$ be replicable and define

$$M_t = \mathbb{E}^{\mathbb{Q}}[e^{-rT} H \mid \mathcal{F}_t].$$

Assume the MRT representation is

$$\mathrm{d}M_t = \psi_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

1. Write the claim price V_t.
2. Find the number of stock shares in the hedge.
3. State the extra property that makes the hedge implementation valid.

Short solution

$$\boxed{V_t = e^{rt} M_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t].}$$

Since $M_t = e^{-rt} V_t = \bar{V}_t$,

$$\mathrm{d}\bar{V}_t = \psi_t\,\mathrm{d}W_t^{\mathbb{Q}} = \Delta_t \sigma \bar{S}_t\,\mathrm{d}W_t^{\mathbb{Q}},$$

so

$$\boxed{\Delta_t = \frac{\psi_t}{\sigma \bar{S}_t}.}$$

The Brownian exposure must be spanned by traded assets, and the strategy must be predictable, self-financing, admissible, and sufficiently integrable.

8.3 Pricing checklist

Eight checks for every martingale-pricing problem

1. What is the numeraire and therefore the discount factor?
2. Under which measure are discounted traded gains martingales?
3. Is the relevant local martingale a true martingale?
4. Is the payoff replicable, or is some risk unspanned?
5. Which Brownian exposures appear in the MRT coefficient?
6. Do the traded assets span those exposures?
7. Which state variables summarize the remaining payoff?
8. Does exercise introduce an optimization over stopping times?

Final synthesis

- Discounted self-financing wealth is a stochastic integral in discounted traded gains.
- Conditional expectation computes the value of a replicable claim once a numeraire and its martingale measure are fixed.
- The MRT reveals Brownian hedge exposures; the asset volatility matrix determines whether those exposures can be traded.
- Completeness is a spanning property. Unspanned risks may leave several martingale measures and ambiguous values for nontraded claims.
- American options replace one terminal expectation by optimal stopping and a Snell envelope.
- Asian options keep the same pricing identity but require an enlarged state or path simulation.

9 Revision Exercises: Lectures 2–6

Four exercises revisit tools needed throughout the course. Each question appears on a single slide with its short answer; the calculations below give the intermediate steps.

9.1 Itô's formula for a nonlinear function

Exercise — the sine of an Itô process

Let $X_0 = 0$ and

$$\mathrm{d}X_t = (1 + X_t)\,\mathrm{d}t + (1 + t)\,\mathrm{d}W_t.$$

Find the stochastic differential of $Y_t = \sin(X_t)$, identifying the quadratic-variation correction.

Short solution

For $f(x) = \sin x$, we have $f'(x) = \cos x$ and $f''(x) = -\sin x$. The finite-variation drift contributes no quadratic variation, while

$$\mathrm{d}[X]_t = (1 + t)^2\,\mathrm{d}t.$$

Itô's formula therefore gives

$$\begin{aligned} \mathrm{d}\sin(X_t) &= f'(X_t)\,\mathrm{d}X_t + \tfrac{1}{2}f''(X_t)\,\mathrm{d}[X]_t \\ &= \boxed{\left[(1+X_t)\cos X_t - \tfrac{1}{2}(1+t)^2 \sin X_t\right]\,\mathrm{d}t + (1+t)\cos X_t\,\mathrm{d}W_t.} \end{aligned}$$

The negative sine term is created by curvature and quadratic variation; it would be absent under the ordinary chain rule.

9.2 Girsanov for a deterministic time-varying drift

Exercise — make the drifted process Brownian

Under $\mathbb{P}$, let $Y_0 = 0$ and

$$\mathrm{d}Y_t = (\alpha + \beta t)\,\mathrm{d}t + \mathrm{d}W_t, \qquad 0 \le t \le T.$$

Construct a density defining an equivalent measure $\mathbb{Q}$ under which Y is Brownian. Explain why the density is a true martingale.

Short solution

Set $\theta_t = \alpha + \beta t$ and define

$$Z_t = \exp\left(-\int_0^t \theta_s\,\mathrm{d}W_s - \tfrac{1}{2}\int_0^t \theta_s^2\,\mathrm{d}s\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

The deterministic drift is square-integrable on every finite horizon:

$$\int_0^T \theta_s^2\,\mathrm{d}s = \alpha^2 T + \alpha\beta T^2 + \frac{\beta^2 T^3}{3} < \infty.$$

Thus Novikov's condition holds and Z is a true martingale. By Girsanov,

$$W_t^{\mathbb{Q}} = W_t + \int_0^t \theta_s\,\mathrm{d}s = W_t + \alpha t + \tfrac{1}{2}\beta t^2$$

is Brownian under $\mathbb{Q}$. Since $Y_0 = 0$, integration of the original SDE gives $Y_t = W_t^{\mathbb{Q}}$, equivalently $\mathrm{d}Y_t = \mathrm{d}W_t^{\mathbb{Q}}$.

9.3 The maximum of Brownian motion with drift

Exercise — Girsanov plus reflection

Let

$$M_T^{\mu} = \max_{0\le s\le T}(W_s + \mu s), \qquad T > 0, \quad a > 0.$$

Use a change of measure and the reflection principle to compute $\mathbb{P}(M_T^{\mu} \ge a)$. Check the result when $\mu = 0$.

Short solution

Set $X_t = W_t + \mu t$ and

$$Z_T = \exp(-\mu W_T - \tfrac{1}{2}\mu^2 T), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Under $\mathbb{Q}$, X is a standard Brownian motion. Rewriting the inverse density in terms of X_T gives

$$\left.\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_T} = \exp(\mu X_T - \tfrac{1}{2}\mu^2 T).$$

Let $\varphi_T(x) = (2\pi T)^{-1/2} e^{-x^2/(2T)}$ and $M_T = \max_{s\le T} X_s$ under $\mathbb{Q}$. For $x < a$, reflection at the first hit of a gives

$$\mathbb{Q}(M_T \ge a,\ X_T \in \mathrm{d}x) = \varphi_T(2a - x)\,\mathrm{d}x.$$

For $x \geq a$, the event $X_T \geq a$ already implies a hit. Therefore

$$\mathbb{P}(M_T^\mu \geq a) = e^{-\mu^2 T/2}\left[\int_a^\infty e^{\mu x}\varphi_T(x)\,\mathrm{d}x + \int_{-\infty}^a e^{\mu x}\varphi_T(2a-x)\,\mathrm{d}x\right].$$

Completing the square in the first integral and substituting $y = 2a - x$ in the second gives

$$\boxed{\mathbb{P}(M_T^\mu \geq a) = \Phi\left(\frac{\mu T - a}{\sqrt{T}}\right) + e^{2\mu a}\Phi\left(\frac{-\mu T - a}{\sqrt{T}}\right).}$$

Here Φ is the standard normal CDF. When $\mu = 0$, the formula reduces to $2[1 - \Phi(a/\sqrt{T})]$, as required by ordinary reflection.

9.4 Stopping times versus times that use future information

Exercise — classify five random times

Use the Brownian filtration $(\mathcal{F}_t)$. Fix $T > 0$ and $a > 0$. Decide which of the following are stopping times:

$$\tau_a = \inf\{t \geq 0 : W_t \geq a\}, \qquad T/2,$$
$$\eta = \inf\{t \geq T/2 : W_t = 0\}, \qquad \rho_T = \sup\{t \leq T : W_t = 0\},$$
$$\gamma_T = \arg\max_{0\leq s\leq T} W_s.$$

Short solution

The first three are stopping times:

- $\{\tau_a \leq t\} = \{\max_{0\leq s\leq t} W_s \geq a\} \in \mathcal{F}_t$;
- $T/2$ is deterministic;
- for $t \geq T/2$, $\{\eta \leq t\}$ says that the observed path has a zero on $[T/2, t]$; for $t < T/2$, the event is empty.

The last zero ρ_T and the time of the maximum γ_T are not stopping times: at a candidate time $t < T$, one must inspect the remaining path to rule out a later zero or a higher value. The criterion throughout is $\{\tau \leq t\} \in \mathcal{F}_t$ for every t.